

Behavior-driven Dropout Prediction Based on User Behavior Data

Final Project
DSI Yan Fang

1. Introduction

Purpose

Employee learning completion is a key driver of workforce capability, compliance, and long-term organizational performance. However, a substantial proportion of employees enrolled in Learning Management System (LMS) courses fail to complete them, resulting in wasted training investment and reduced learning outcomes.

The purpose of this study is to develop a robust and interpretable machine learning model to predict course dropouts at the enrollment level.

Learning Management System (LMS) Dataset

The dataset used in this study is derived from an enterprise-scale LMS operating in the Human Capital Management (HCM) domain[1]. It captures user enrollments, learning behaviors, and demographic context across multiple data sources.

Core data include enrollment records (user–course mappings), detailed event logs that track user interactions with learning objects (e.g., video views, problem attempts, navigation events), and demographic attributes such as education level and geographic information. These datasets collectively provide a comprehensive view of learner engagement, behavior patterns, and contextual factors throughout the course lifecycle.[1]

Target Variable and Features

The LMS dropout dataset is formulated as a binary classification problem. The target variable is `dropout_flg` (1 = dropout, where 1 indicates course dropout and 0 indicates course completion. The dataset is defined at the enrollment level (user–course pairs).

Features are constructed by aggregating user learning behaviors from event logs, including interactions with Video, Wiki, Problem, Discussion, Navigation, and Access objects, together with demographic and geographic attributes. Behavioral features capture engagement intensity and learning patterns over time. The final feature set consists of behavior-driven features complemented by user background information, while excluding post-outcome variables to ensure model validity

Recent research

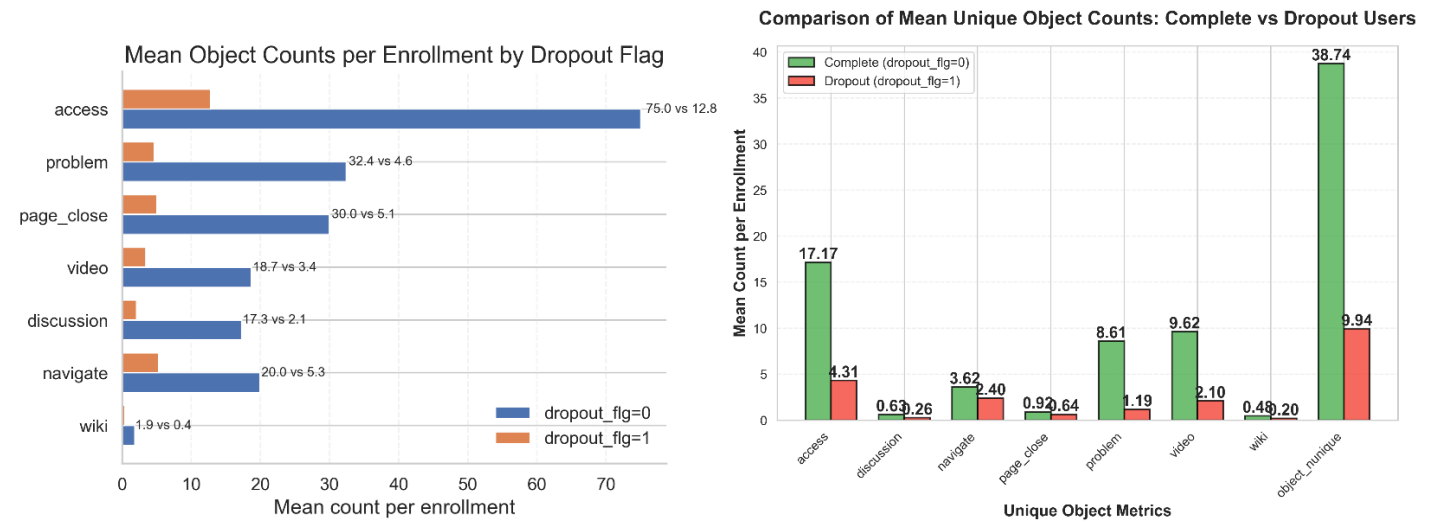
Recent research on learner dropout prediction has extensively leveraged large-scale MOOC datasets released by XuetangX. [2][3] These studies aim to improve predictive modeling of course completion outcomes using enrollment information, fine-grained activity logs, and demographic attributes. Prior work demonstrates that behavioral features derived from learning logs—such as video views, problem attempts, and access frequency—are strong indicators of student persistence and dropout risk.

Several studies apply classical machine learning and neural models to XuetangX data and report competitive performance in predicting dropout. For example, [2]Feng et al. analyze dropout patterns using XuetangX-MOOCs and highlight the importance of temporal engagement behaviors. [3] Xu et al. further show that combining behavioral and demographic features improves predictive accuracy in dropout classification tasks. More recent work explores advanced settings such as federated learning to model dropout across distributed educational data.[4]

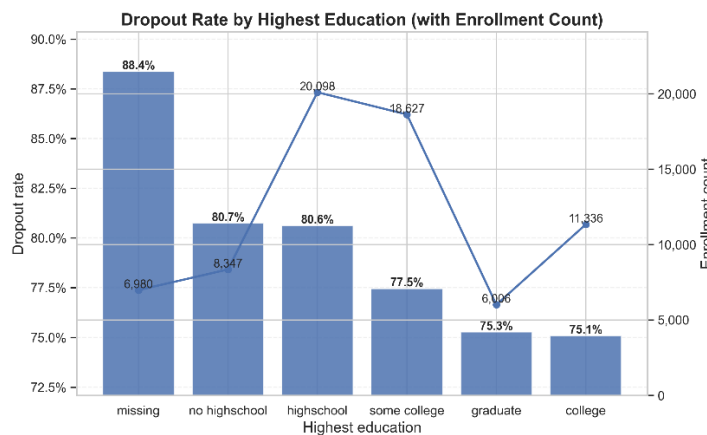
2. Exploratory Data Analysis

I used the describe function to check for missing values as well as the minimum and maximum values in the dataset, and generated histograms, distribution plots, violin plots, boxplots for all features to obtain an overall understanding of feature scale and distribution.

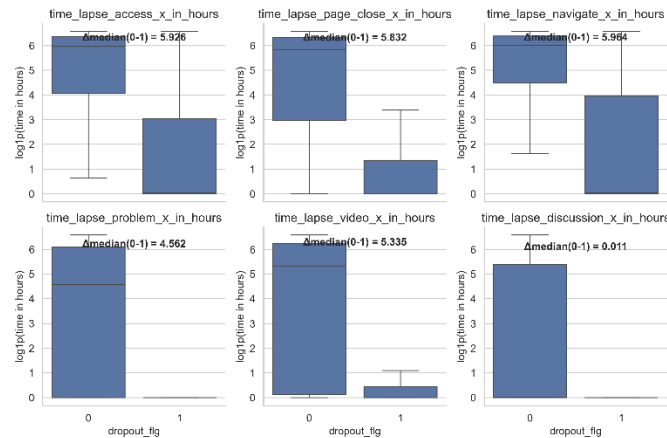
The **object count** (the total number of times a user triggers a given event type) and **unique object count** (the number of distinct objects a user interacts with under that event type) indicate that non-dropout users engage more frequently overall. In contrast, dropout users exhibit a lower average number of unique objects, suggesting reduced engagement with course materials.



The bar–line chart shows users with missing or lower educational attainment exhibit higher dropout rates. In parallel, enrollment counts vary substantially across categories, with high school and some college groups contributing the largest user volumes. The box plots-across all selected features, non-dropout users exhibit substantially higher median engagement levels. Importantly, the observed differences are driven by systematic shifts in the median rather than by extreme outliers, indicating broad behavioral differences instead of noise from a small number of highly active users.



<The total number of courses taken and the dropout rate of users with different levels of education >

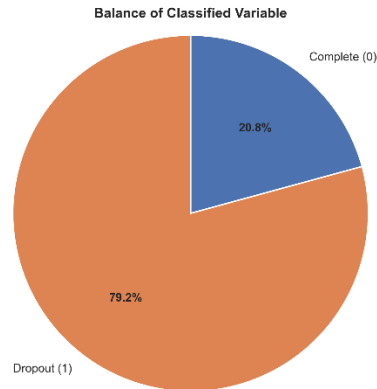


<Distributions of time-lapse engagement features by dropout status.>

Distributions of log-transformed time-lapse engagement features by dropout status. Dropout cases consistently show lower median engagement times across activities, suggesting a strong negative association between time investment and persistence.

Target Variable

The target variable in this dataset is drop_out. Each user-data is classified to two groups: '0' or '1', which stands for completer and dropout.



<The difference between dropouts and completers>

3. Methods

Data Preprocessing

Prior to model training, features were standardized using **StandardScaler** to ensure that all variables were on the same scale. This preprocessing step prevents features with larger magnitudes from dominating the learning process and does not impose any artificial bounds on the data distribution.

Dataset Split

The dataset was first split into **training and test sets(80: 20)** using a stratified strategy to preserve the class distribution of the target variable (dropout_flg). The held-out test set was reserved exclusively for final model evaluation and was not

involved in model selection or hyperparameter tuning.

For Logistic Regression, KNN, Random Forest, and XGBoost models, Stratified K-Fold cross-validation (5 folds) was applied on the training set. This cross-validation scheme implicitly serves as the validation mechanism, providing stable and unbiased performance estimates while efficiently utilizing the available data. Cross-validation was also used within hyperparameter tuning procedures to select optimal model configurations.

Features

The dataset contains both continuous and categorical features with missing values. For continuous features, missing values were handled through an imputation procedure applied separately to the training and test sets to avoid data leakage. For categorical features, missing values were treated as a new category “missing”

ML Pipeline

Four machine learning models were evaluated: **Logistic Regression, KNN, Random Forest, and XGBoost**. To address class imbalance, SMOTE was incorporated into the training pipelines for Logistic Regression and KNN, while Random Forest and XGBoost handled imbalance through class weighting strategies.

Model	CV Mean ROC AUC	CV Std ROC AUC	Test ROC AUC	Test F1	Test Recall	Test Precision
Logistic Regression	0.849211	0.001866	0.838298	0.844710	0.853941	0.843850
KNN	0.662735	0.005418	0.672040	0.712251	0.690571	0.750767
Random Forest	0.857020	0.004465	0.852113	0.853820	0.865371	0.857477
XGBoost	0.847780	0.004555	0.849957	0.840700	0.836517	0.846859

Given the imbalanced class distribution (approximately 2:8 for classes 0 and 1) and the fact that class 1 represents the primary target of interest, model evaluation prioritizes the ability to correctly identify class 1 instances. In this context, I prioritize F2-score as the primary metric to emphasize recall under class imbalance, while also reporting precision and F1 for balance analysis. KNN exhibited substantially lower cross-validation and test performance compared to the other models. Given its consistently inferior performance and known limitations in high-dimensional settings, further hyperparameter tuning for KNN was not pursued.

Logistic Regression	C: [0.01, 0.1, 1, 10, 100] Solver: ['liblinear', 'saga'], Penalty:[L1,L2], max_iter: [2000, 5000] class_weight: [None, 'balanced']
Random Forest	n_estimators: [100, 200, 300], max_depth: [10, 20, None] min_samples_split': [2, 5] min_samples_leaf: [1, 2] max_features: ['sqrt', 'log2'] class_weight: ['balanced', 'None']
XGboost	n_estimators: [100, 200, 300] max_depth: [3, 5, 7,] learning_rate: [0.01,0.1, 0.2] colsample_bytree: [0.8, 1.0],

	subsample: [0.8, 1.0], 'min_child_weight': [1, 3] scale_pos_weight: [1, scale_pos_weight]
--	---

After hyperparameter tuning via **RandomizedSearchCV**, the best-performing model for each algorithm was evaluated on the independent test set using ROC AUC, F1 score, precision, and recall.

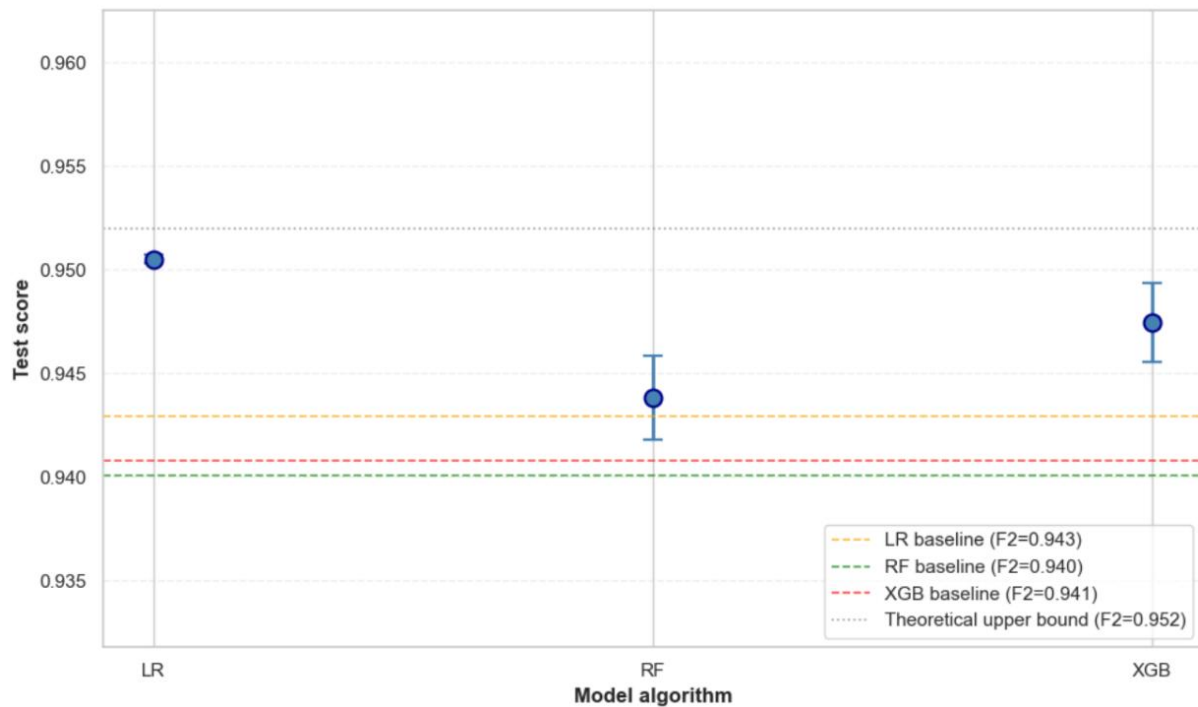
A 5-fold Stratified K-Fold cross-validation strategy with shuffling was employed on the training data for all models. This approach maintains consistent class proportions across folds while reducing variance in performance estimation. Cross-validation scores were used for model selection and hyperparameter optimization, whereas the test set remained untouched until the final evaluation stage. The resulting cross-validation performance showed minimal variability across folds, reflecting the robustness of the training procedure.

4. Result

	Mean test score	Standard Deviation	(Mean test score-baseline)/standard deviation
Logistic Regression	0.9505	0.0002	-6.99
Random Forest	0.9438	0.0020	-4.06
XGboost	0.9474	0.0019	-2.40

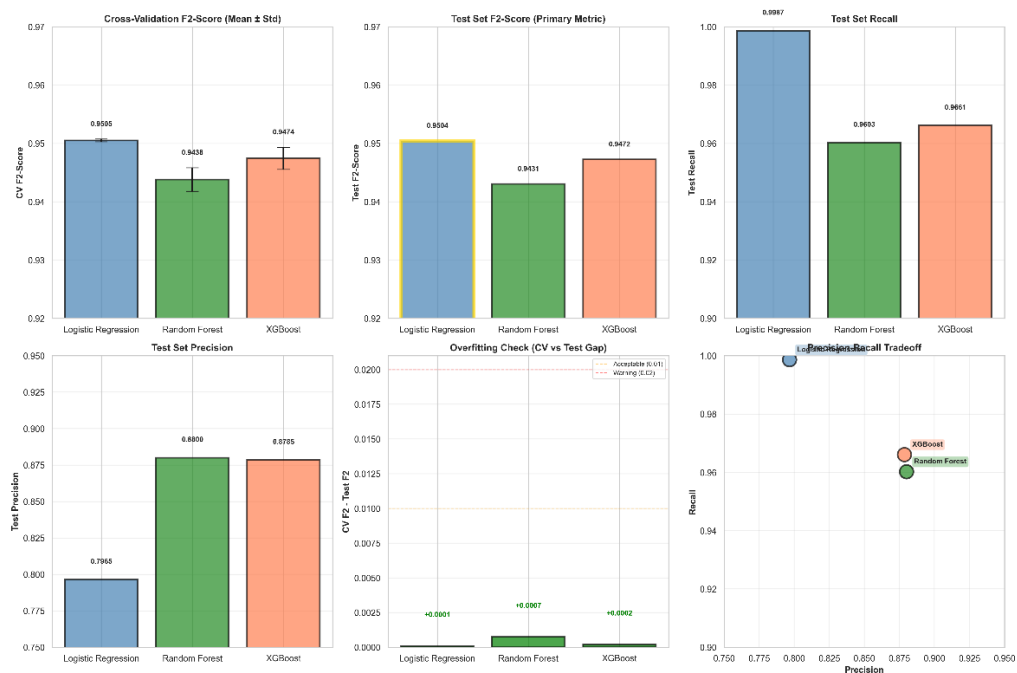
<Mean Value of each Machine Learning Algorithm >

The negative standardized improvement indicates that the majority-class baseline outperforms the learned models under the F2 metric. This is expected given the strong class imbalance and the asymmetric nature of F2, which heavily prioritizes recall. In this setting, a trivial strategy that always predicts the positive class achieves near-optimal F2 performance, despite being impractical and having very low precision. Therefore, the negative standardized improvement does not imply that the learned models are ineffective, but rather highlights the limitation of F2 as a sole evaluation metric under severe class imbalance.



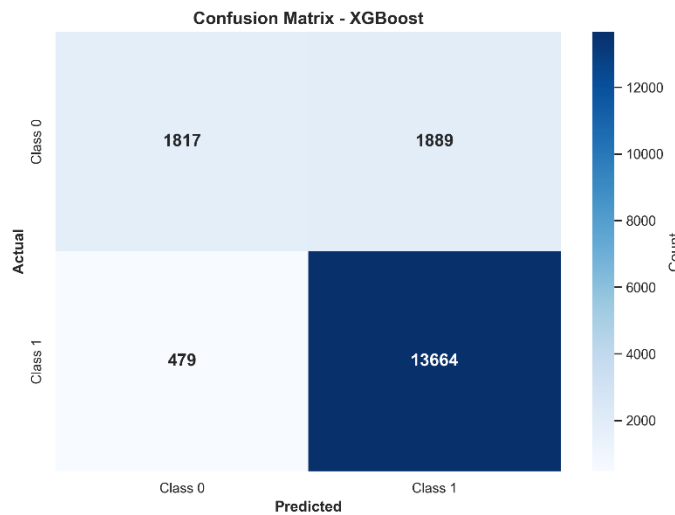
<Test Score F2 Compare with baselines>

I additionally compare the tuned models with their default (untuned) counterparts as engineering baselines. While none of the models surpass the theoretical majority-class baseline under the F2 metric, all tuned models consistently outperform their default versions, indicating that hyperparameter tuning yields meaningful and stable improvements. This comparison confirms that the observed performance gap is primarily driven by the metric's bias under class imbalance, rather than ineffective model learning.

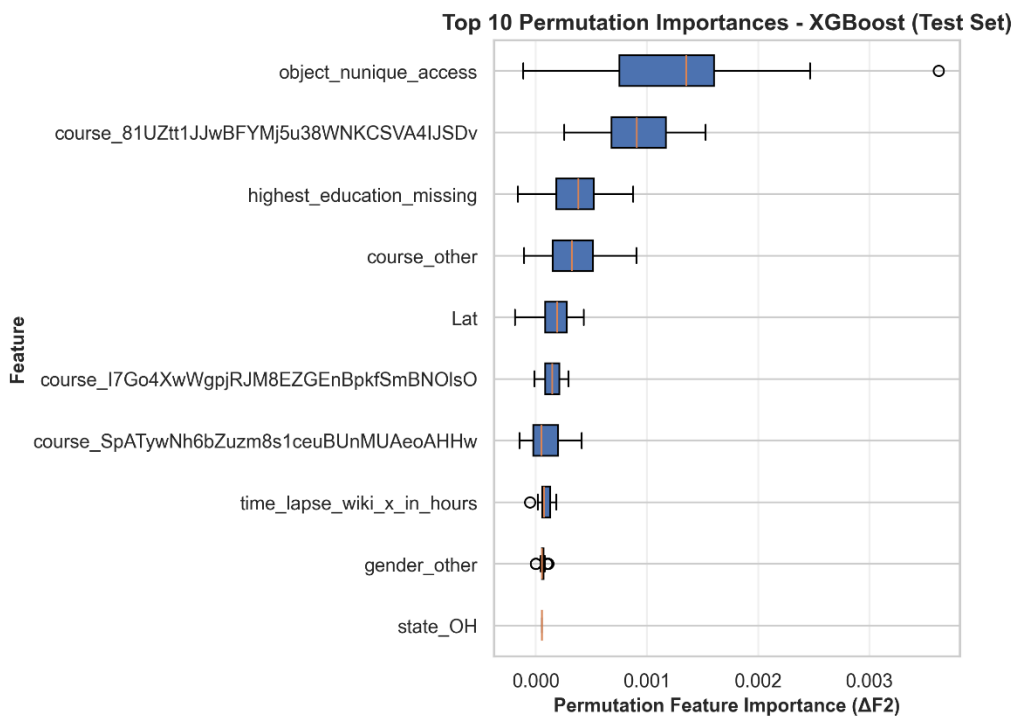


<balance of F2, Recall and Precision>

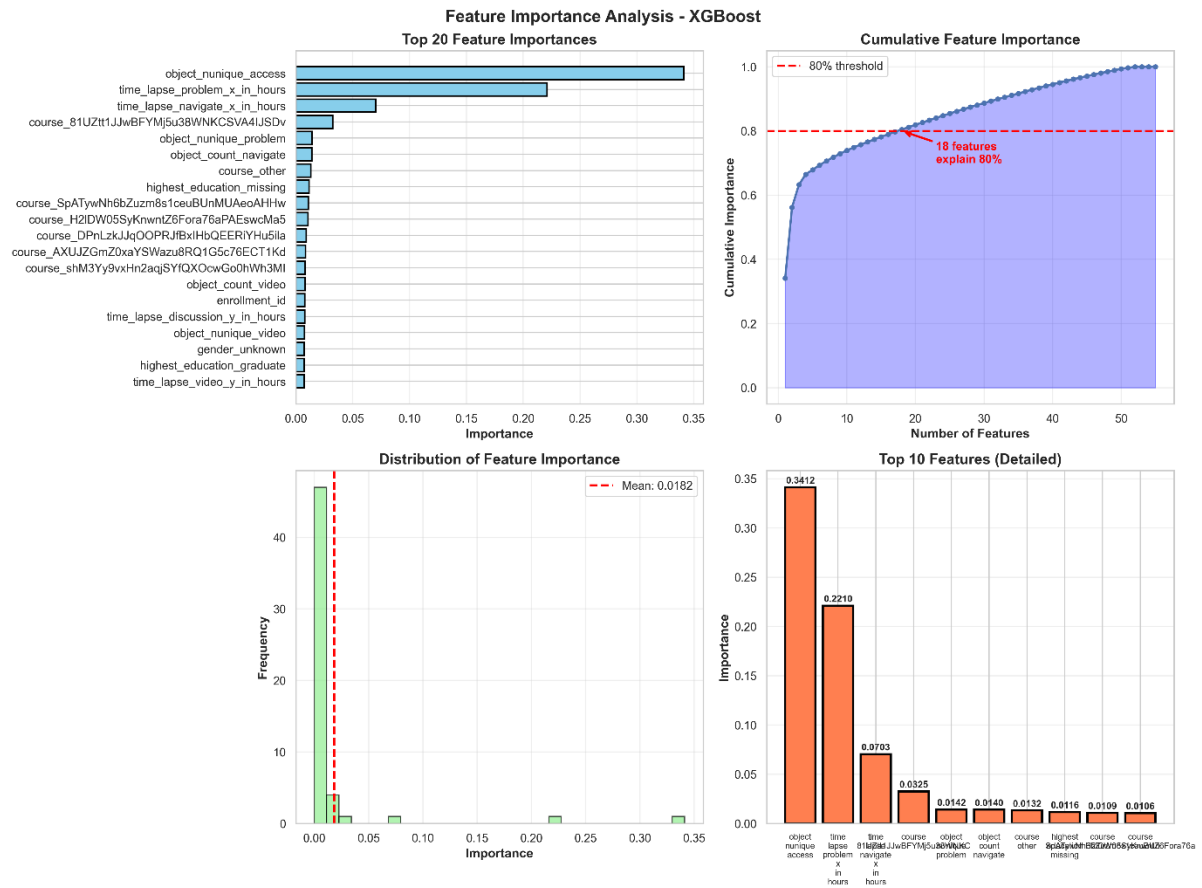
While Logistic Regression achieves the highest recall and is therefore optimal for recall-driven objectives, it does so at the cost of substantially lower precision. In contrast, XGBoost provides a more balanced trade-off between recall and precision, resulting in stronger F2 performance and a more stable overall decision boundary. Therefore, when the objective is to balance recall with precision rather than maximize recall alone, XGBoost is selected as the preferred model.



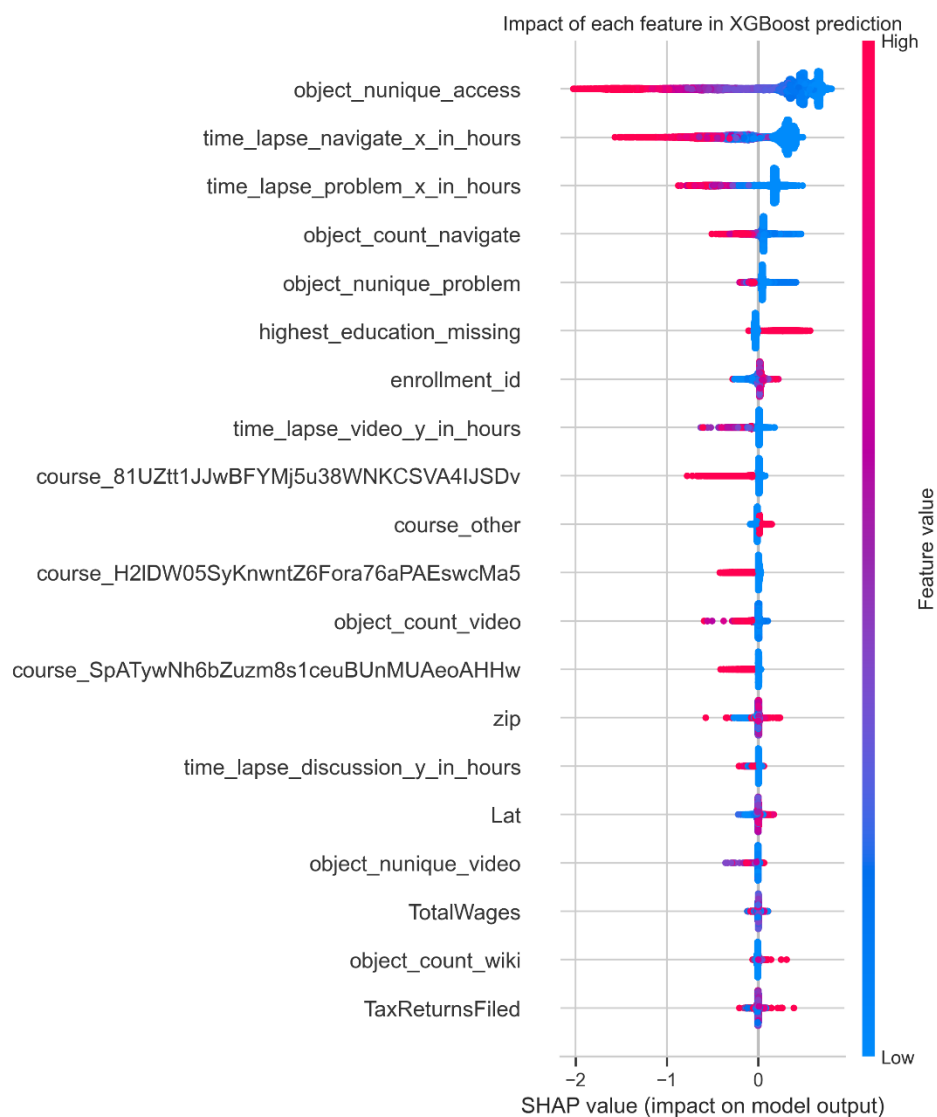
<Test set confusion matrix using the XGboost Model>



<Top 10 Permutation Importances Using XGboost test set >



<Feature importance analysis of the XGBoost model, including top-ranked features, cumulative importance, and the overall distribution of importance scores.>

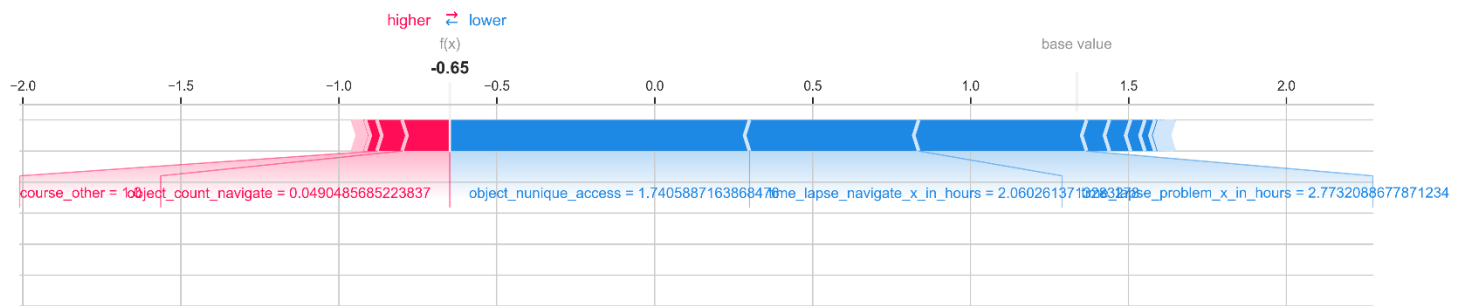


<Impact of each feature in XGboost Prediction>

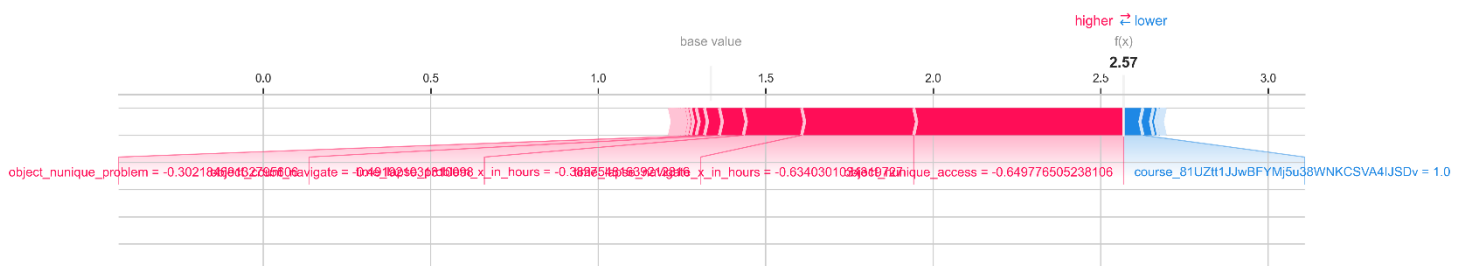
Across XGBoost built-in feature importance, permutation importance on the test set, and SHAP value analysis, a consistent set of engagement-related features—such as unique content access and time spent on navigation and problem-solving activities—emerge as the primary drivers of model predictions. These features exhibit stable importance rankings and coherent directional effects, with lower engagement systematically increasing the predicted probability of dropout. The agreement among multiple interpretability methods indicates that the models' decision logic is robust, generalizable, and not driven by spurious correlations. Therefore, while the majority-class baseline achieves near-optimal performance on the F2 metric, the trained models provide substantial added value by uncovering actionable behavioral signals that are critical for understanding and monitoring student dropout risk.

The predictions of XGBoost are mainly driven by "learning behavior intensity + learning rhythm", rather than demographic or static background features. The model can distinguish between "high participation but low risk" users and "low participation high risk" users, which is precisely why it outperforms Logistic Regression in the

precision-recall balance.



<Shap local Value for index 5>



<Shap local Value for index 10>

For this instance, several features exert strong downward pressure on the prediction, pushing the model toward class 0. `course_other = 1.0` (attend in a course) produces a large negative SHAP value, showing strong local influence despite low global importance. `object_nunique_access` (The number of different objects that the user has accessed in the access event) and `time_lapse_navigate_x_in_hours` (time spent in navigation) further reinforce the class 0 outcome by contributing additional negative impact. Although a few features (e.g., `object_nunique_problem` (The number of different objects that the user has accessed in the pproblem event)) push slightly toward class 1, their effects are minor and insufficient to offset the dominant negative contributors.

5. Outlook

Although XGBoost achieves a strong balance between recall and precision, there is still room for improvement in robustness and generalization. Future work could optimize the decision threshold using cost-sensitive or validation-based criteria to better align predictions with downstream decision costs, and ensemble approaches combining linear and tree-based models may further improve performance. A key limitation of the current approach is its reliance on behavioral features within a fixed time window; incorporating temporal or longitudinal features could better capture evolving user behavior. Finally, collecting richer contextual data (e.g., user intent or course difficulty) and validating the model on external datasets would help improve generalization and reduce overfitting.

6. References

[1] XuetangX MOOC Dataset. *User Enrollment and Learning Activity Data*.

MoocData Repository: <https://www.moocdata.cn>

[2] Feng, W., Tang, J., & Liu, X. (2019), *Understanding Dropouts in MOOCs*.

Proceedings of the AAAI Conference on Artificial Intelligence (AAAI).

[3] Zhang, T., Liu, H., Tao, J., Wang, Y., Yu, M., Chen, H., & Yu, G. (2023). *Enhancing Dropout Prediction in Distributed Educational Data Using Learning Pattern Awareness: A Federated Learning Approach*.

Mathematics, 11(24), 4977.

<https://doi.org/10.3390/math11244977>

[4] Hong, B., Wei, Z., & Yang, Y. (2017). *Discovering Learning Behavior Patterns to Predict Dropout in MOOC*.

Proceedings of the International Conference on Computer Science & Education (ICCSE), 700–704.

<https://doi.org/10.1109/ICCSE.2017.8085583>

7.Github Repository

<https://github.com/yan259479/Dropout-classification.git>